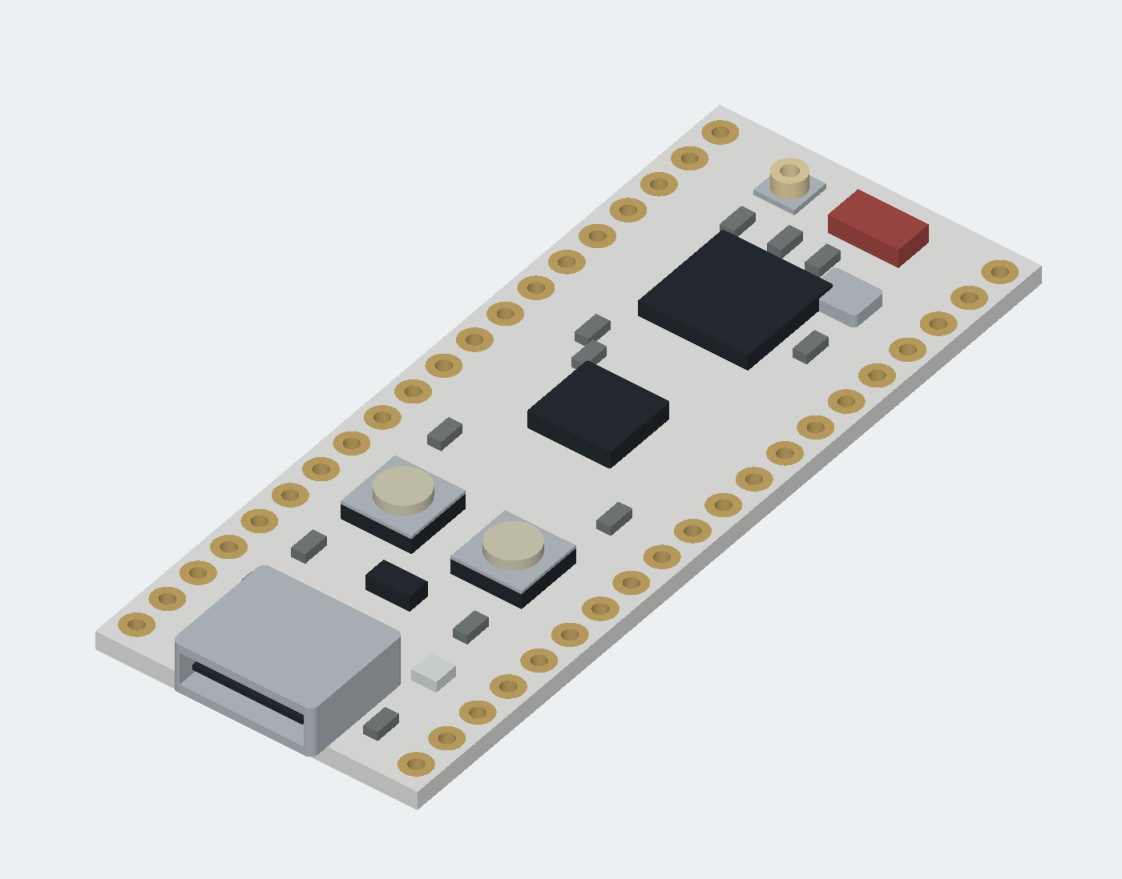
ESP32-S3-N16R8

Development board - mechanical model and electrical reference

This document describes the narrow white PCB in the supplied photographs, not a generic Espressif DevKitC-1. Manufacturer, revision and schematic are unknown. Dimensions explicitly shown in the supplied drawing are distinguished from estimated heights and component envelopes.

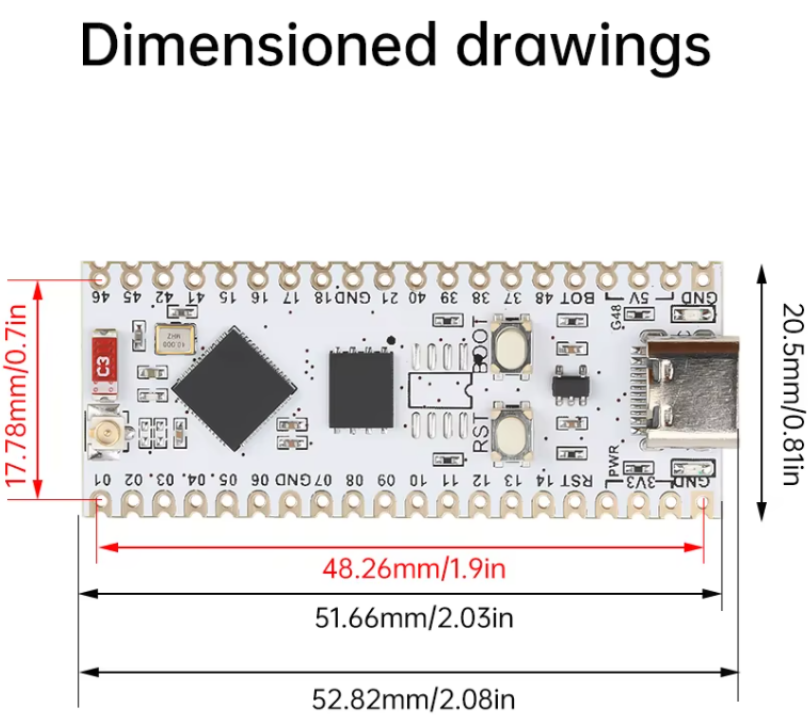


Bare board model, component side. USB-C is at the near end; external RF connector is at the far end.

Item	Reference value / scope
Board and connector envelope	20.50 wide x 52.82 long; estimated bare thickness envelope 4.40
Header layout	2 rows of 20; pitch 2.54; row centers 17.78 apart; first-to-last span 48.26
Requested configurations	Bare board and downward-facing male headers. STEP, LYL, 3MF, STL and previews included.
Memory naming	N16R8 convention: 16 MB flash, 8 MB PSRAM. This board's installed parts must be verified. [S2]

Critical discrepancy: R14 is marked 37 in the photos, but 47 in the supplied pinout. Its electrical routing remains unverified. GPIO37 is not spare I/O when used by octal PSRAM. [S1, S3]

Mechanical dimensions and height estimates



Source drawing: 17.78 mm row spacing is already dimensioned in red; no photo-only guess was needed.

Feature	Value (mm)	Evidence / interpretation
PCB outline	20.50 x 51.60	Dimensioned drawing; rectangular reference outline
USB-inclusive length	52.82	Drawing; 1.22 connector overhang beyond one PCB end
Header pitch / span	2.54 / 48.26	User's pitch; 19 intervals x 2.54 = 48.26; drawing agrees
Row spacing / centers	17.78 / X +/-8.89	Drawing and photo proportions; 7 x 2.54
End / side margins	1.67 / 1.36	(51.60 - 48.26)/2; (20.50 - 17.78)/2
Hole / plated pad diameter	1.00 / 1.90	Estimated; fit-check physical header pins
PCB thickness	1.20	Estimated; underside datum is elevation 0
USB shell W x L x H	9.00 x 7.60 x 3.20	Estimated; shell sits on PCB top at 1.20
Bare maximum elevation	4.40	Estimated PCB 1.20 + USB shell 3.20
MCU / flash / crystal height	1.00 / 1.10 / 0.80	Estimated above PCB top; not component datasheet dimensions
Header plastic / exposed tail	2.54 / 6.00	Assumed generic downward 1x20 male header
Square pin / top solder stub	0.64 / 0.20	Assumed; pin tip -8.54, solder tip +1.40
With-header overall height	12.94	Estimated: max +4.40 minus min -8.54

Coordinate convention: component-side up, antenna at -Z and USB at +Z in Layerling. X runs left-to-right. Hole centers: X = +/-8.89; Z = -24.13 + 2.54 i, i = 0...19. STEP is Z-up (STEP Y is opposite Layerling Z). PCB underside is the height datum.

All 40 header positions

View from the component side, antenna at top and USB at bottom. L01/R01 are at the antenna end; L20/R20 are at the USB end. These are reference IDs assigned for this model, not manufacturer pin numbers. The supplied photo determines placement; chip alternate functions are checked against [S1, S3].

Row	Left signal	Left function / caution	Right signal	Right function / caution
01	GPIO1	ADC1 CH0; Touch1	GPIO46	reset-time strap; download-mode condition
02	GPIO2	ADC1 CH1; Touch2	GPIO45	reset-time strap; VDD_SPI selection
03	GPIO3	ADC1 CH2; Touch3; reset-time strap	GPIO42	JTAG MTMS
04	GPIO4	ADC1 CH3; Touch4	GPIO41	JTAG MTDI
05	GPIO5	ADC1 CH4; Touch5	GPIO15	ADC2 CH4
06	GPIO6	ADC1 CH5; Touch6	GPIO16	ADC2 CH5
07	GND	Ground; common signal return	GPIO17	ADC2 CH6
08	GPIO7	ADC1 CH6; Touch7	GPIO18	ADC2 CH7
09	GPIO8	ADC1 CH7; Touch8	GND	Ground; common signal return
10	GPIO9	ADC1 CH8; Touch9	GPIO21	General-purpose digital I/O
11	GPIO10	ADC1 CH9; Touch10	GPIO40	JTAG MTDO
12	GPIO11	ADC2 CH0; Touch11	GPIO39	JTAG MTCK
13	GPIO12	ADC2 CH1; Touch12	GPIO38	General-purpose digital I/O
14	GPIO13	ADC2 CH2; Touch13	37 / 47 UNVERIFIED	**Photo 37 / chart 47: UNVERIFIED**; avoid until routing checked
15	GPIO14	ADC2 CH3; Touch14	GPIO48	RGB LED claimed by supplied chart
16	RST / EN	CHIP_PU / EN; active-low reset	BOOT / GPIO0	GPIO0 boot strap; normally high
17	3V3	3.3 V rail; current capability unknown	5V	5 V / USB supply rail; isolation unknown
18	3V3	3.3 V rail; current capability unknown	5V	5 V / USB supply rail; isolation unknown
19	3V3	3.3 V rail; current capability unknown	5V	5 V / USB supply rail; isolation unknown
20	GND	Ground; common signal return	GND	Ground; common signal return

GPIO19 (USB D-) and GPIO20 (USB D+) are not exposed in these header rows. GPIO43/44 UART0 are also not shown; UART signals can be routed through the GPIO matrix in software. GPIO22-25 do not exist on ESP32-S3. No schematic has established the three repeated power pins' exact connections.

Chip capabilities and electrical cautions

Capability	Verified ESP32-S3 family information
CPU / internal RAM	Dual-core Xtensa LX7 up to 240 MHz; 512 KB internal SRAM. [S4]
Radio	2.4 GHz Wi-Fi 802.11 b/g/n; Bluetooth 5 LE. Not Bluetooth Classic; not 5 GHz Wi-Fi. [S4]
External memory	N16R8 naming means 16 MB flash / 8 MB PSRAM in Espressif module nomenclature; ESP32-S3R8 embeds 8 MB octal-SPI PSRAM. This PCB is not identified as a WROOM module. [S1, S2]
Interfaces	GPIO matrix, I2C, SPI, UART, I2S, PWM, RMT, ADC, touch, SD/MMC and USB. Peripheral routing is flexible but not unrestricted. [S3, S4]
Analog	ADC1: GPIO1-10; ADC2: GPIO11-20. GPIO19/20 overlap USB. 14 capacitive-touch inputs on GPIO1-14. [S1, S3]
USB	Full-speed USB (12 Mbit/s), HID/CDC and composite devices using TinyUSB. Two USB controllers share one internal PHY. [S5]

3.3 V logic - Do not apply 5 V to GPIO or 3V3. Chip supply recommendations are 3.0-3.6 V, not a claim that the board's 3V3 regulator tolerates arbitrary input. [S1]

Power rails - 5V header pins likely connect to the USB supply path, but regulator, reverse-current protection and output current are unknown. Use one known supply at a time; do not back-power USB from an external 5V source without verifying the circuit.

Strapping - GPIO0, GPIO3, GPIO45 and GPIO46 are sampled around reset. GPIO45 can select flash/PSRAM supply voltage if eFuses do not override it. Pulling it high may break boot on 3.3 V memory designs. GPIO46 must be low with GPIO0 low for joint download boot. [S1]

PSRAM reservation - GPIO33-37 are involved in octal memory; GPIO37 must not be repurposed on an R8 configuration using it. Treat ambiguous R14 as unavailable until verified. [S1, S3]

ADC and Wi-Fi - The supplier warns against simultaneous ADC2 and Wi-Fi. Current ESP-IDF oneshot documentation describes arbitration/protection, not an unconditional ban. Prefer ADC1 for joystick signals, use the proper driver and calibration, and handle conversion errors. [S6]

LED and JTAG - GPIO48 is marked as the RGB LED in this board's supplied chart (not GPIO38 from another board). GPIO39-42 overlap external JTAG; configure intentionally. Exact LED protocol and polarity are unknown.

USB, startup and controller integration

Native USB - The chip USB signals are GPIO19 D- and GPIO20 D+. The single pictured USB-C connector is consistent with native USB, but routing is not confirmed by a schematic. Test enumeration before designing firmware around this assumption. HID mouse/custom/composite devices are supported by Espressif's TinyUSB integration; a SpaceMouse-compatible protocol still requires firmware. [S5]

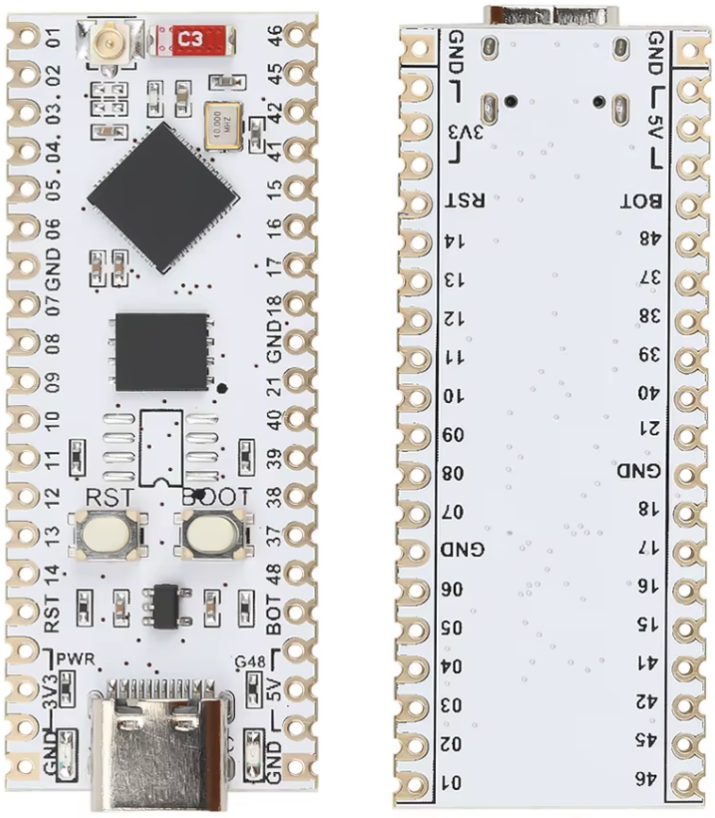
Recovery boot - Hold BOOT, press and release RST, then release BOOT and attempt flashing over the appropriate USB interface. This assumes BOOT drives GPIO0 and RST drives EN as labeled; GPIO46 must meet its download-mode strap condition. Use a USB data cable. [S1]

Memory verification - Read the actual flash ID/capacity with esptool and verify detected PSRAM in ESP-IDF or Arduino firmware before selecting a 16 MB partition table or octal-PSRAM settings. Report successful initialization and measured sizes; do not infer correctness from the listing alone.

Joystick / display / buttons - Prefer non-strapping ADC1 GPIO4-10 for Hall sensor signals. Digital display and button signals can use verified general-purpose pins; reserve USB19/20, avoid ambiguous R14 and consider the LED/JTAG conflicts. Level-shift incompatible signals and check display supply requirements.

Mechanical installation - Allow a cable plug and bend radius beyond the 52.82 mm board envelope. Provide access to BOOT/RST, relief for the RF plug/cable, mating-header height and cooling. The model has no antenna, socket board, cable plug, fasteners or validated mounting bosses.

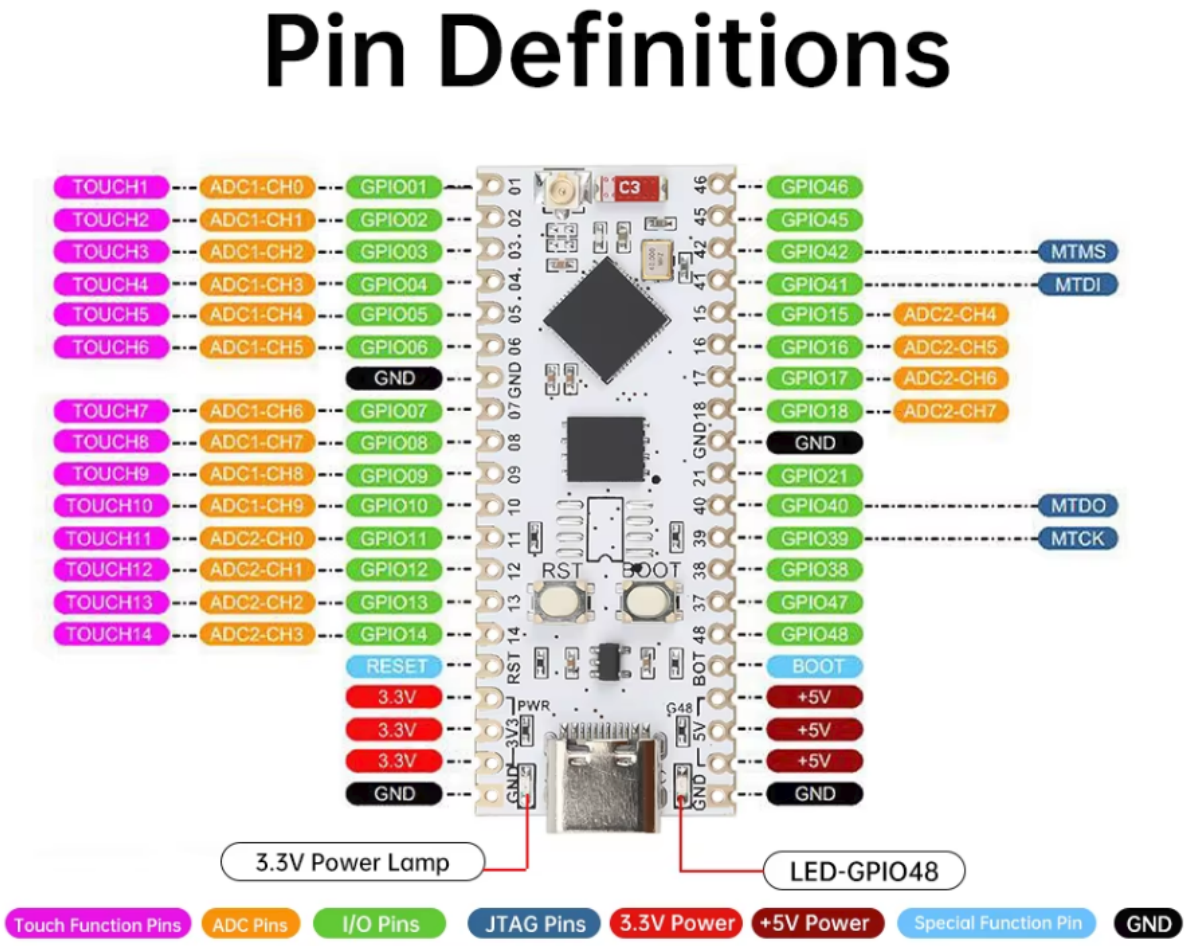
RF / power integrity - Use a compatible external antenna and verify its connector on the sample. Keep metal and high-current switching away from the antenna/cable; provide adequate regulated supply and local decoupling. This document does not establish antenna certification or board-level EMC performance.



Supplied front/back photograph; underside is rotated in the source image. Pinout on page 3 is normalized to component-side orientation, not read directly from the rotated back view.

Supplied pin-definition graphic

Retained for provenance, not accepted wholesale as a verified schematic. Use the corrected table on page 3.



Discrepancies: the graphic says GPIO47 where both photos mark 37; routing is unknown. Its blanket GPIO45/GPIO46 pull-resistor warning is replaced by the specific boot/voltage strap cautions on page 4. Its ADC2/Wi-Fi claim is qualified using current ESP-IDF documentation. GPIO46 is listed as I/O/T in the current chip datasheet, not categorically input-only. [S1, S6]

Sources, assumptions and verification boundary

Research date: **1 November 2026**. This is an original reference summary. Chip-family documentation supports capabilities and pin cautions, but does not prove this seller board's routing, regulator or revision. No exact board-specific schematic or manufacturer manual was identified.

[S1] Espressif ESP32-S3 Series Datasheet, v2.2
Chip variants, GPIO functions, boot straps, supply recommendations and package reference.
https://www.espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf

[S2] Espressif ESP32-S3-WROOM-1 / WROOM-1U Datasheet, v1.8
N16R8 memory nomenclature only; this model is not claimed to contain that shielded module.
https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

[S3] ESP-IDF ESP32-S3 GPIO and RTC GPIO reference
ADC pin map, GPIO matrix, memory-reserved pins, USB19/20 and strapping summary.
https://docs.espressif.com/projects/esp-idf/en/stable/esp32s3/api-reference/peripherals/gpio.html

[S4] Espressif ESP32-S3 product overview
CPU, SRAM, Wi-Fi, Bluetooth LE and supported interfaces.
https://www.espressif.com/en/products/socs/esp32-s3

[S5] Espressif ESP-USB USB Device Stack for ESP32-S3
USB D-/D+ routing, full-speed operation, TinyUSB classes and shared PHY limitation.
https://docs.espressif.com/projects/esp-usb/en/latest/esp32s3/usb_device.html

[S6] ESP-IDF ESP32-S3 ADC Oneshot Mode Driver
ADC2/Wi-Fi sharing and driver protection; calibration and error handling.
https://docs.espressif.com/projects/esp-idf/en/stable/esp32s3/api-reference/peripherals/adc/adc_oneshot.html

Supplied local references

ESP32-S3-N16R8_dimensions.png; ESP32-S3-N16R8_front_back.png; ESP32-S3-N16R8_pins.png. Copies are included alongside the models. These supplied images establish the visual identity, dimensioned outline and provisional header labeling.

What remains approximate

Every undocumented height; drill/pad size; component positions and packages except the nominal reference layout; pin/header lengths; exact USB and RF receptacle geometry; solder and tracks. Edge scallops/castellations and an unpopulated landing pattern are omitted. Model colors/materials are illustrative. Exact STEP preserves modeled solids, not proof of physical manufacturing tolerances.

Before production

Measure PCB thickness, connector heights, header fit and antenna mating; establish R14 routing; verify power paths and actual flash/PSRAM; exercise ADC and USB; compare enclosure clearances with a physical sample. Model validation checks the modeled envelopes and valid solids, not an unmeasured board.